The Entrepreneurial State and the Public Origins of Private Innovation

Introduction

In my previous paper, on Elizabeth Anderson's *Private Government*, I argued that the authority employers exercise over workers is constituted by the State: the relation liberal doctrine describes as private is established and sustained in public law. This paper extends the argument from authority to wealth. Following Mariana Mazzucato's *The Entrepreneurial State*, it argues that the State does more than create the conditions for economic growth. At the points where innovation is most uncertain it is often the principal investor, and much of what is celebrated as private invention rests on publicly funded foundations. The first part examines the conventional view, in which the market is an equilibrium system, innovation arises spontaneously from private enterprise, and the State's proper role is confined to correcting market failures. The second reconstructs Mazzucato's alternative, drawing on Karl Polanyi's account of the economy as an instituted process: innovation, like the market itself, is made rather than found, and the State is among its makers. The third turns to the question Mazzucato's argument ultimately raises, which is not who creates innovation but who is rewarded for it.

I. The Conventional View: Innovation as a Spontaneous Development of the Market

The conventional picture of the market economy begins from a definition. Economics, on Lionel Robbins's influential formulation, studies human behavior as a relationship between given ends and scarce means that have alternative uses.[1] The market is the institution that solves this problem at the level of society: agents pursuing their own ends respond to prices, prices adjust to balance supply and demand, and the result, under idealized conditions, is an allocation of resources that cannot be improved for anyone without making someone else worse off. The model is powerful, but it is static. It describes the conditions of equilibrium rather than the process by which an economy changes, and it has little to say about the most conspicuous fact of modern capitalist economies, which is that they do not stay still.

Joseph Schumpeter made this fact the center of his account. "Capitalism, then, is by nature a form or method of economic change and not only never is but never can be stationary."[2] What drives the change is innovation, a "process of industrial mutation—if I may use that biological term—that incessantly revolutionizes the economic structure from within, incessantly destroying the old one, incessantly creating a new one. This process of Creative Destruction is the essential fact about capitalism."[3] Schumpeter corrects the equilibrium picture by making disruption, rather than balance, the normal condition of the economy. But he preserves one of its assumptions. The agent of creative destruction is private: the entrepreneur who introduces new goods, methods, and forms of organization, and, by the time of *Capitalism, Socialism and Democracy*, increasingly the research department of the large corporation. The State has no comparable role in his account of innovation.

The conventional view does assign the State a role, and that role needs stating accurately, since it is what Mazzucato argues against. Its classic statements are Richard Nelson's and Kenneth Arrow's.[4] Knowledge, they argued, has the properties of a public good: once produced it can be used by many at no addi-

[1] Lionel Robbins, *An Essay on the Nature and Significance of Economic Science* (London: Macmillan, 1932).

[2] Joseph A. Schumpeter, *Capitalism, Socialism and Democracy* (1942; New York: Harper Perennial Modern Thought, 2008), 82.

[3] Schumpeter, 83.

[4] Richard R. Nelson, "The Simple Economics of Basic Scientific Research," *Journal of Political Economy* 67, no. 3 (1959): 297–306; Kenneth J. Arrow, "Economic Welfare and the

tional cost, and its producer cannot fully prevent others from using it. Firms unable to capture the full return on basic research will invest less in it than would be socially optimal, and the remedy is public funding of basic science, together with the wider infrastructure private research requires. In Dani Rodrik's summary, the State "provides the essential social, legal, and educational infrastructure that sustains private R&D."[5] Beyond this, on the market-failure view, the State should step aside. It should correct the specific failures that justify its intervention and otherwise leave the direction of innovation to private actors, above all resisting the temptation to "pick winners."

Mazzucato's objection is that this account misdescribes what States have actually done. The State, she argues, has driven innovation "not by simply fixing market failures but by actively creating and shaping (new) markets, while regulating existing ones."[6] The claim is not that the market-failure rationale is wrong, but that it is too modest to explain the historical record, and that its modesty has political consequences. If the State only fixes failures, then the value produced by innovation belongs, in principle, to the private actors who bring it to market.

II. Innovation as an Instituted Process: The State as Lead Investor

Mazzucato's argument draws on a tradition that denies the market is a natural order which the State either respects or disturbs. Polanyi's essay "The Economy as Instituted Process" begins by distinguishing two meanings of "economic."[7] The formal meaning derives from the logic of choice under scarcity; it is Robbins's economizing, the allocation of scarce means among ends. The substantive meaning refers to the ways human beings actually secure their livelihood through interaction with nature and with one another. In the substantive sense the economy is always instituted, embedded in and organized by institutions of

Allocation of Resources for Invention," in *The Rate and Direction of Inventive Activity: Economic and Social Factors* (Princeton, NJ: Princeton University Press, 1962), 609–626.

[5] Dani Rodrik, "Democratizing Innovation," Project Syndicate, August 12, 2020, https://www.project-syndicate.org/commentary/policymakers-should-influence-course-of-technological-innovation-by-dani-rodrik-2020-08.

[6] Mariana Mazzucato, *The Entrepreneurial State: Debunking Public vs. Private Sector Myths*, Rev. ed. (2013; New York: PublicAffairs, 2018), 6.

[7] Karl Polanyi, "The Economy as Instituted Process," in *The Sociology of Economic Life*, ed. Mark Granovetter and Richard Swedberg (1957; Boulder, CO: Westview Press, 1992), 29–51.

reciprocity, redistribution, or exchange. The market is one such institution, not the natural condition of economic life. The question is therefore not whether the State intervenes in a market that would otherwise exist on its own, but how a market economy is instituted, and by whom.

Polanyi's answer, in *The Great Transformation*, is that the self-regulating market of the nineteenth century was built by the State. "The road to the free market was opened and kept open by an enormous increase in continuous, centrally organized and controlled interventionism."[8] And in a sentence that inverts the usual story: "While laissez-faire economy was the product of deliberate State action, subsequent restrictions on laissez-faire started in a spontaneous way. Laissez-faire was planned; planning was not."[9] What appears spontaneous was constructed, and what appears as interference was the unplanned reaction of a society protecting itself from the market's effects. Fred Block, whose introduction accompanies the edition of Polanyi cited here, extended the analysis to the contemporary United States. Beneath its market-fundamentalist rhetoric, Block argued, the American state has operated since the 1980s as a "hidden developmental state," funding and brokering networks of research through agencies such as DARPA, the National Institutes of Health, and the Small Business Innovation Research program.[10] Mazzucato's work builds directly on this finding.

Her central claim concerns the kind of risk innovation involves. Frank Knight distinguished risk, whose probabilities can be estimated and so priced and insured, from uncertainty, which cannot be.[11] Innovation at the frontier is uncertain in Knight's sense. No one knows in advance whether a line of research will succeed, what it will produce, or when, and the time horizons involved often exceed what private investors will tolerate. Venture capital, the institution usually credited with financing innovation, typically enters only after the most uncertain phase is over, when a technology is proven enough to promise returns within a few years. The earlier and harder stage has frequently been financed by the State. This is why Mazzucato calls the State a lead investor rather than a market-fixer, and why she returns repeatedly to Keynes's state-

[8] Karl Polanyi, *The Great Transformation: The Political and Economic Origins of Our Time*, 2nd ed., with an introduction by Fred Block, with a foreword by Joseph E. Stiglitz (1944; Boston: Beacon Press, 2001), 146.

[9] Polanyi, 147.

[10] Fred Block, "Swimming against the Current: The Rise of a Hidden Developmental State in the United States," *Politics & Society* 36, no. 2 (2008): 169–206.

[11] Frank H. Knight, *Risk, Uncertainty and Profit* (Boston: Houghton Mifflin, 1921).

ment of the State's proper task: "The important thing for Government is not to do things which individuals are doing already, and to do them a little better or a little worse; but to do those things which at present are not done at all."[12]

Her most discussed illustration is the iPhone. Nearly every technology that makes the device possible originated in, or was substantially funded by, public research: the internet, which grew out of the Defense Department's ARPANET; GPS, developed by the military; the touchscreen; and the voice assistant Siri, which descends from a DARPA-funded artificial intelligence project.[13] Apple's contribution was real, but it was one of integration, design, and marketing, built on a foundation of publicly funded invention. The popular image of innovation as the work of a solitary genius, whether Schumpeter's entrepreneur or his contemporary incarnation in the technology founder, obscures the collective and largely public labor on which the celebrated product depends.

This dependence compounds over time. W. Brian Arthur has argued that technologies are built out of other technologies: each new technology combines existing ones, so the stock of available technologies grows by recombining itself.[14] In his words, "[technological] change breeds further change endogenously and continually."[15] Public investment matters so much in this process because it seeds the combinatorial base. A technology developed with public funds becomes an element that private firms can recombine into products, which in turn create demand for further technologies and make still others possible. The State's early investments are not isolated interventions but contributions to a cumulative process whose later and more visible stages are credited to the private sector.

III. Risks and Rewards: From Symbiosis to Parasitism

If the State bears a large share of the uncertainty, the question becomes who receives the returns. Here Mazzucato's argument turns critical, and it is easy

[12] John Maynard Keynes, *The End of Laissez-Faire* (London: Hogarth Press, 1926).

[13] Mazzucato, *The Entrepreneurial State*, chap. 5.

[14] W. Brian Arthur, *The Nature of Technology: What It Is and How It Evolves* (New York: Free Press, 2009).

[15] W. Brian Arthur, *Complexity Economics: A Different Framework for Economic Thought*, SFI Working Paper (Santa Fe, NM: Santa Fe Institute, 2013), https://sites.santafe.edu/~wbarthur/Papers/Comp.Econ.SFI.pdf.

to misread. She does not claim that the relationship between the public and private sectors is symbiotic. She claims that it ought to be, and that in its current form it is parasitic: the risks of innovation are socialized while the rewards are privatized.[16] The public sector absorbs the losses on the investments that fail and captures little of the gains from those that succeed.

Two American clean-energy investments make the point concrete. In 2009 and 2010 the Department of Energy's loan programs lent $535 million to the solar manufacturer Solyndra and $465 million to Tesla. Solyndra went bankrupt in 2011, and the loss became a political scandal. Tesla survived and repaid its loan early, in 2013, and the public received its interest and nothing more, despite having financed the company at a moment when private lenders would not.[17] A venture capitalist expects most investments to fail and makes its return on the few that succeed spectacularly. In this arrangement the State takes a venture capitalist's losses without its upside.

The aggregate figures tell a similar story once disaggregated. Private firms fund roughly seventy percent of American research and development and perform somewhat more.[18] But private spending is concentrated in development, the later and less uncertain end of the process, while the public share is far larger at the basic and early-stage end. The private majority rests on a public foundation. And because private investment decisions are concentrated in a relatively small number of firms and investors, the direction of innovation reflects their priorities. As Rodrik observes, this "often leads them to under-invest in technologies that have significant long-run returns, such as those that reduce climate change, or to pay inadequate attention to the human-rights or privacy implications of digital innovations."[19] Innovation is too consequential to be directed solely by those who profit from it.

Mazzucato's argument has drawn serious criticism, and it should be met rather than ignored. Deirdre McCloskey and Alberto Mingardi contend that she conflates funding research with entrepreneurship: to finance the science behind a component is not to discover what consumers will want, or to bear the commercial risk of bringing a product to market.[20] They argue that her

[16] Mazzucato, *The Entrepreneurial State*, chap. 8.

[17] Mazzucato, ??

[18] National Center for Science and Engineering Statistics, *National Patterns of R&D Resources*, Data series (Alexandria, VA: National Science Foundation).

[19] Rodrik, "Democratizing Innovation."

[20] Deirdre Nansen McCloskey and Alberto Mingardi, *The Myth of the Entrepreneurial State* (Great Barrington, MA: American Institute for Economic Research, 2020).

examples are selected for success, that the State's failures receive less attention than its triumphs, and that much of what she credits to the entrepreneurial state was military spending pursued for strategic rather than economic ends. They also argue that the public already recovers its investment through the taxes that commercial success generates. The first objection has force. Apple's integration of public technologies into a product millions wanted was a genuine achievement that no government agency performed. But the objection does not touch the central claim about rewards. Even if the private integrators deserve substantial credit, the distribution of returns does not track the distribution of risk. The tax argument is the strongest reply to that claim, and it is weaker than it looks. Taxation returns to the public a share of the profits of every firm, whether or not public investment made those profits possible, so it is not a return on the particular risks the State bore; and the firms that have gained most from publicly funded research are often among the most effective at avoiding tax, a point Mazzucato presses against Apple itself.[21] The critics establish that the private sector contributes; they do not establish that it should capture nearly all of the value that public investment made possible.

Mazzucato's remedies follow from this diagnosis. She proposes that the State retain equity stakes in the firms it supports, issue loans whose repayment is contingent on income, attach conditions to public funding, and direct the returns into funds that finance the next round of public investment.[22] The aim is to let the State share in the upside of risks it already bears, so that its successes can pay for its failures in the way a venture capitalist's do.

Conclusion

Read together with Anderson, Mazzucato completes an argument about the word "private." Anderson shows that the authority of the firm is constituted by the State; Mazzucato shows that much of the wealth of the innovative firm is seeded by it. In both cases, what liberal doctrine presents as the spontaneous product of private action turns out to be an arrangement the State makes, and the question of who benefits from it is political rather than natural.

The argument nonetheless leaves a difficulty unresolved. To call the private capture of publicly funded innovation "parasitic" is to claim that the public sector created value that others appropriated, and that claim requires a theory of

[21] Mazzucato, *The Entrepreneurial State*, chap. 8.

[22] Mazzucato, chap. 9.

value. If value is simply what the market pays, as the conventional view holds, then whoever receives the returns has by definition created the value, and there is nothing to call parasitic. Mazzucato needs an account of collective value creation that does not reduce to market prices, and *The Entrepreneurial State* assumes such an account more than it provides one; she develops it only later, in *The Value of Everything*.[23] Her remedies sharpen the difficulty. An equity stake measures the State's contribution by the returns the market assigns to the firm, which is the very measure whose inadequacy the diagnosis depends on. Whether the entrepreneurial state can claim a share of what it creates without adopting the calculus of the capital it finances is the question the book leaves open.

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